Cointegration Tests Summary — Week 7 Part 2

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At a glance

Even a spread that looks stationary may be a coincidence. Part 2 decides whether a genuine long-run equilibrium exists using two formal tests. It comes down to three points.

1. **Engle-Granger is a two-step residual test.** Regress one level on the other to get β, then test whether the residuals are stationary. Because the residuals are estimated, it uses its own critical values rather than the ordinary ADF ones.
2. **Johansen tests the whole system.** It fits a VECM and reads the cointegrating rank, that is, how many independent equilibria there are. It treats every variable symmetrically and suits three or more series.
3. **Prove it before you trade it.** Check that each series is I(1), use the right test and critical values, and check robustness. A positive result puts β and the spread in your hands.

1. Starting point: from pictures to tests

Part 1 gave the concept of cointegration; Part 2 decides it statistically. A spread seen on a chart is not evidence.

1. **We can now recognise cointegration.** Two I(1) series whose combination is stationary share a long-run equilibrium, and the spread is the anchor. But a picture alone proves nothing.
2. **Eyeballing is not enough.** In a short sample a spread can look mean-reverting by chance. We need a formal test with proper critical values.
3. **Two standard tests do the job.** Engle-Granger for a pair, Johansen for a system. Both ask the same question rigorously: "is there a stationary combination?"

Step 0: first check that each series is I(1)

Cointegration is a statement about I(1) series. So before testing, first confirm that each series is "non-stationary in levels, stationary in differences".

$$X_t \sim I(1) \iff \text{ADF on } X_t:\ \text{not reject}, \quad \text{ADF on } \Delta X_t:\ \text{reject}$$

- The null hypothesis of the ADF (Augmented Dickey-Fuller) test is "there is a unit root (non-stationary)". For I(1), it must fail to reject in levels and reject in differences. It is safer to look alongside a test with the opposite null, such as KPSS.
- If a series is already stationary (I(0)), cointegration is not the right tool.
- If a series is I(2), needing two differences to become stationary, the set-up itself changes.

Getting the order of integration right is step 0 of any cointegration test.

Two branches of testing

	Engle-Granger	Johansen
Approach	Two steps, single equation	System approach based on the VECM
Procedure	Regress one series on the other, then test the residuals	Handles many series at once and finds **how many** equilibria there are
Strengths	Simple and intuitive. Suited to a single pair	More powerful and symmetric, but more complex

2. The Engle-Granger two-step test

Engle-Granger takes just two steps: "get β by regression, then see whether the residuals are stationary". The verdict comes from the residuals in step 2.

Step 1: estimate the long-run relation

Regress the level of one series on the level of the other by OLS.

$$\text{Step 1:} \quad Y_t = \alpha + \beta X_t + u_t \xrightarrow{\text{OLS}} \hat{u}_t = Y_t - \hat{\alpha} - \hat{\beta} X_t$$

- The slope $\hat{\beta}$ estimates the cointegrating coefficient, and the residual $\hat{u}$ is the estimated equilibrium error (the spread).
- This $\hat{\beta}$ is the hedge ratio used for trading in Part 3.
- As in the lecture figure, β is the slope of the regression line through the scatter of the two levels. If the series are cointegrated, this slope is the weight that makes the spread stationary.

Step 2: test the residuals for stationarity

Run an ADF test on the step 1 residuals. It regresses the change in the residuals on their previous value.

$$\text{Step 2:} \quad \Delta \hat{u}_t = \gamma \, \hat{u}_{t-1} + \sum_{i=1}^{p} \phi_i \, \Delta \hat{u}_{t-i} + e_t, \qquad H_0 : \gamma = 0 \;\; \text{vs.} \;\; H_1 : \gamma < 0$$

- If γ = 0, the residuals are a random walk (unit root). The null hypothesis is "**not** cointegrated".
- If γ < 0, the residuals are pulled back towards 0 whenever they stray. That is, they are stationary and the series are cointegrated.
- The lagged difference terms (φ) absorb autocorrelation in the residuals. The number of lags p is chosen by an information criterion.

$$t_\gamma = \frac{\hat{\gamma}}{\mathrm{se}(\hat{\gamma})} < c_{\mathrm{EG}} \Rightarrow \text{reject } H_0 \Rightarrow \text{cointegrated}$$

Stationary residuals certify a real relationship; wandering residuals mean the regression was spurious. Everything rests on this one stationarity test.

Why separate critical values are needed

The step 2 residuals are not observed values but **estimates**. So the ordinary ADF critical values must not be used.

- **The problem**: OLS chooses β to make the residuals as small as possible. So the residuals are biased towards looking more stationary than they are. With ordinary ADF critical values, the null is rejected too often and false cointegration is found.
- **The fix**: use the Engle-Granger (MacKinnon) critical values. They correct for the estimation of β and are stricter.

5% critical value (with constant)	Value
Ordinary ADF (observed series)	About −2.86
Engle-Granger, two variables (estimated residuals)	About −3.34

※ These are asymptotic critical values from MacKinnon's tables. They become more negative as the number of variables grows. Software (such as coint in statsmodels) applies them automatically; just do not run an ordinary ADF on the residuals yourself and use its p-value.

Which variable is the dependent one

- The β from regressing Y on X and the β from regressing X on Y are not reciprocals of each other. In finite samples the test conclusion can differ too. The method is not symmetric in the two variables.
- What to do: try both orderings and check that the conclusions agree. If one variable is more exogenous, use it as the regressor. Or avoid the issue altogether with Johansen.

Limits of Engle-Granger

- It can find **at most one** cointegrating relation.
- With three or more series there may be several relations, but it cannot tell you **how many** there are.
- The result depends on which variable is put on the left-hand side.

What is needed is a system test that treats every series symmetrically and reports the number of relations. That is the Johansen approach.

3. The Johansen test and the cointegrating rank

Johansen fits a vector error-correction model (VECM) directly and asks how many independent stationary combinations exist among the series. It treats every variable symmetrically.

Where Johansen looks: the rank of the matrix Π

Write the system in a form that "explains the changes (ΔY) by the previous levels (Y_{t−1}) and past changes". It is the Week 6 VAR rearranged in differences.

$$\Delta \mathbf{Y}_t = \Pi\, \mathbf{Y}_{t-1} + \sum_{i=1}^{p-1} \Gamma_i\, \Delta \mathbf{Y}_{t-i} + \boldsymbol{\varepsilon}_t, \qquad r = \operatorname{rank}(\Pi)$$

- ΔY: the vector of changes (returns) of each series. Γ is the coefficient matrix holding the short-run dynamics.
- **ΠY_{t−1}** term carries the equilibrium information. It is the channel through which the previous levels affect the next changes.
- The **rank** of Π (the number of linearly independent rows) is the number of cointegrating relations.

With rank r, Π splits into the product of two matrices. This is the form used in Part 3.

$$\Pi = \underbrace{\alpha}_{k\times r}\ \underbrace{\beta^{\top}}_{r\times k} \qquad (\beta : \text{cointegrating vectors}, \quad \alpha : \text{adjustment speeds})$$

- Each column of β is a cointegrating vector. β^⊤Y are the stationary spreads.
- α is how quickly each series reacts to a deviation of that spread and moves back (the adjustment speed).

Reading the cointegrating rank r

With k series, r runs from 0 to k.

$$r = 0 : \ \text{no cointegration} \qquad 0 < r < k : \ r \text{ relations} \qquad r = k : \ \text{all } I(0)$$

Rank	Meaning
r = 0	Π = 0. Nothing ties the levels — no cointegration. A VAR in differences is the right model
0 < r < k	Exactly r long-run relations. There are r tradable equilibria
r = k	Π has full rank. The series were all stationary to

Rank	Meaning
	begin with

With k I(1) series there can be at most k − 1 cointegrating relations. For example, r = 2 with four series means two independent equilibria. Johansen finds this number; Engle-Granger cannot.

Trace and maximum-eigenvalue statistics

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^{k} \ln\left(1 - \hat{\lambda}_i\right), \qquad \lambda_{\max}(r) = -T \ln\left(1 - \hat{\lambda}_{r+1}\right)$$

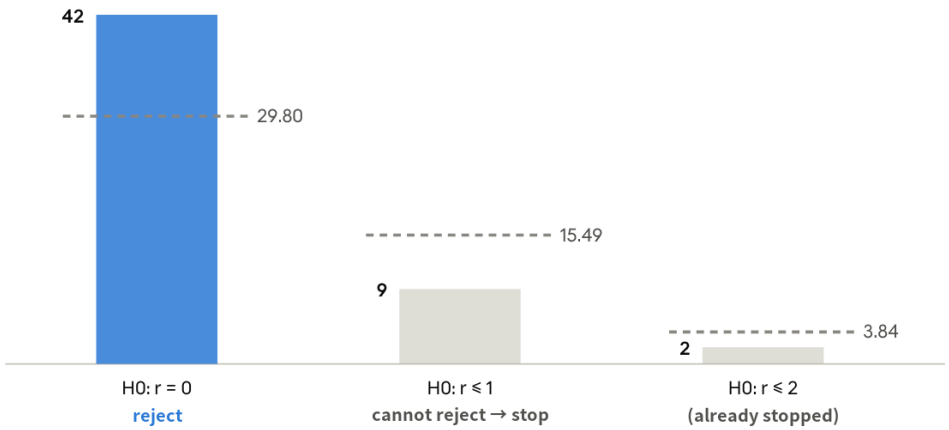
- $\hat{\lambda}_i$ are the eigenvalues obtained from Π, sorted from largest. Each lies between 0 and 1 and shows how strongly stationary the combination in that direction is. Close to 0 means there is no equilibrium in that direction.
- **Trace test**: tests "at most r relations" against "more than r". It pools the information in all remaining directions (from r+1 to k). It is the form most often reported.
- **Max-eigenvalue test**: tests "exactly r" against "exactly r+1". It focuses on the next single relation.
- The two usually agree. If they differ, dig into why.

Procedure for setting the rank: step down until you fail to reject

1. Test the null r = 0. If it is rejected, there is at least one relation.
2. Then test r ≤ 1, r ≤ 2 … in turn. At each step compare the statistic with the critical value, and reject if the statistic is larger.
3. Stop at the first null you cannot reject. That r is the rank.

Only r = 0 is rejected, so the cointegration rank is 1

Johansen trace statistics for 3 series (bars) vs 5% critical values (dashed) · reject when a bar passes its line



Statistics from the lecture figure (about 42, 9, 2) and standard 5% trace critical values (with constant)

At the first step the statistic 42 exceeds the critical value 29.80, so r = 0 is rejected. At the second step 9 falls short of 15.49, so r ≤ 1 cannot be rejected; we stop here, and the rank is 1.

4. Which test to use

Engle-Granger is a two-step test for quickly checking a pair; Johansen is a symmetric system test that finds how many equilibria there are. Choose according to the problem.

Feature	Engle-Granger	Johansen
Number of series	2 (a pair)	2 or more
Approach	Single equation, two steps	System (VECM-based)
Relations found	At most 1	At most k − 1 (the rank)
Symmetric in variables	No (you pick the dependent variable)	Yes
Output	One β and the spread	All cointegrating vectors and adjustment speeds

Use Engle-Granger for a quick pair check, and Johansen when there are many series or the number of equilibria matters. For a single pair, the two tests should broadly agree, and the Johansen rank should be 0 or 1.

5. Practical pitfalls and workflow

Pitfall	Good practice
Structural breaks can hide cointegration or fake it	First check each series is I(1), and check for structural breaks
Short samples have low power and miss real relations	Use enough data and be suspicious of borderline results
The lag length of the test changes the result	Choose lags by an information criterion and check robustness

Five steps to a defensible verdict

1. **Order of integration**: run unit-root tests on each series to confirm all are I(1).
2. **Choose the test**: Engle-Granger for a pair, Johansen for more.
3. **Run it**: use the right critical values (EG or Johansen).
4. **Interpret the result**: cointegrated or not? How many relations?
5. **Check robustness**: vary the lags, the ordering of variables and the subsample, and look for structural breaks.

A positive result gives β and the spread. These are the raw material for error correction and trading in Part 3.

6. Common misconceptions and self-check

Common misconception	Reality
A spread that looks stationary proves cointegration	In short samples appearances deceive. Run a formal test (Section 1)
Ordinary ADF critical values are fine in step 2	The residuals are estimates, so use Engle-Granger critical values (Section 2)
Engle-Granger and Johansen are interchangeable	EG suits a pair; Johansen suits a system and counts the rank (Section 4)

Three questions before moving on

1. **What exactly does step 2 of Engle-Granger test?** Whether the step 1 regression residuals are stationary (have no unit root).
2. **With four series, the Johansen rank comes out as 2. What does that mean?** There are two independent long-run equilibria (cointegrating relations).
3. **Why must ordinary ADF critical values not be used in step 2?** Because the residuals are estimated by OLS and are therefore biased towards stationarity.

Next: Part 3

Error correction: the force that restores equilibrium, the VECM and the adjustment speed α, and pairs trading built on the cointegrated spread.